

Tomographic Coherence-Based Neural Rendering

*A Falsifiable Neuroscience Protocol with Firewalled Bridges to Pattern, Action, Consciousness
Science, and Physics*

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Abstract

A brain preparing a grip does not receive one finished description of its own state. Different electrodes, cells, and moments expose partial measurements. The scientific problem is whether relations among those partial measurements contain a reproducible, task-relevant structure that a downstream system could use, beyond what is already available in voltage, power, firing rate, evoked response, temporal history, or an equally capable model of the raw signal. The proposed solution is to estimate those relations, test whether they improve held-out prediction, destroy them selectively with matched nulls, and reserve causal language for a later intervention. Only after this problem, operation, and test are stated do we give the proposal its name.

Tomographic Coherence-Based Neural Rendering (TCNR) proposes that distributed, partial measurements can be stitched into a task-relevant state estimate partly through structured phase relations. This paper has two explicitly firewalled layers. Part I turns one bounded neural claim into a prospective protocol using the public Brochier et al. instructed-delay reach-to-grasp recordings from two macaques. It asks whether beta-band spatial phase organization during the delay improves held-out prediction of the cue-instructed grip condition under the declared representation classes, compared with evoked activity, spike-rate summaries, amplitude, power, scalar coherence, non-oscillatory temporal dynamics, an unengineered 1-kHz LFP comparator, and an equal-compute nonlinear LFP comparator. One animal is used for development; all feature definitions, model-selection rules, nulls, and decision thresholds are then locked before the second animal is evaluated. The primary estimand is the cross-fitted binary log-loss improvement of the phase-structured model over the strongest locked comparator, with 0.02 bits / trial as the smallest effect of interest. Phase scrambling, block-preserving label permutation, circular shifting, coordinate permutation, evoked-response controls, and a reference / phase-estimator sensitivity set test whether any gain is explained by power, common input, geometry, filtering, or the declared model class. In the source task, the fixed visual cue specifies PG versus SG; cue identity and grip instruction are therefore perfectly confounded. This dataset cannot identify a maintained motor plan independently of cue identity. This is a protocol paper. It reports no dataset result and does not test consciousness, causal necessity, or literal tomography.

Part II develops a conceptual bridge from relational differences to higher-order patterns, prediction, action, and feedback. It distinguishes statistical dependence, neural oscillatory coherence, computational integration, and quantum coherence; places TCNR beside current consciousness-science comparators; and asks what a physical implementation would need to show before cross-scale language becomes more than analogy. The physics discussion is conditional and local: it does not infer quantum coherence from neural synchrony, identify spacetime with a computer, or treat coherence as sufficient for mind. It also distinguishes a physical difference from a difference available through a particular measurement channel, treats hidden-source recovery as an identifiability problem, and separates classical synchronization from quantum entanglement. The combined endpoint is a theory-and-methods paper. It reports no dataset result, consciousness measurement, cross-substrate validation, or physical unification.

Scope and evidence boundary

Locally final-preprint-ready theory-and-methods article; no new empirical result

1. Claim, Scope, And The Empirical Firewall

In ordinary language, the central question is whether the *arrangement* of activity across the array helps identify the instructed condition after the best available descriptions of the activity itself have already been supplied. Existing neuroscience provides signal processing, decoding, traveling-wave analysis, communication-through-coherence hypotheses, latent-state models, and causal perturbation methods. TCNR adds a particular joined operation: partial measurements are related in space and time, the relation is used to estimate a task-relevant latent state, that estimate is compared with equal-input alternatives, and a later perturbation asks whether the relation changes receiver transfer. “Tomographic” is therefore earned only when a forward operator and reconstruction test are identified; otherwise the narrower phrase is *relational recurrent reconstruction*.

TCNR names the full theory-and-methods program. The frozen empirical model label **M6-TNR** is retained from the preregistration and denotes the narrower tomographic neural-rendering feature model tested in Stage A; it should not be read as the complete TCNR program.

The Stage A claim is:

During the instructed delay, engineered beta-band spatial phase features improve out-of-sample prediction of the cue-instructed precision-versus-side grip condition under the declared representation classes, relative to evoked activity, spike / rate summaries, amplitude, power, scalar coherence, temporal history, an unengineered 1-kHz LFP comparator, and an equal-compute nonlinear LFP comparator.

The target is an experimentally defined cue-instructed task condition, not a private percept. The first fixed LED cue directly specifies PG versus SG in this dataset, so cue identity and instructed grip are not separately identifiable. Evoked-response controls can test whether a declared feature class improves prediction beyond a cue-locked waveform comparator; they cannot make the motor-plan interpretation independent of cue identity. A positive result would support only one bounded TNR component: engineered phase structure has useful held-out predictive bias under the declared model classes. It would not establish new information absent from the LFP, phase necessity, a literal tomographic inversion, a cue-independent maintained plan, or consciousness.

The protocol separates three questions:

1. **Representation utility:** do phase-geometry features improve held-out prediction under the declared comparator classes?
2. **Mechanistic specificity:** does the improvement survive controls that preserve power, autocorrelation, common drive, or raw input?
3. **Causation:** would selective phase perturbation change receiver transfer at matched energy?

Stage A addresses the first two observationally. Stage B states a prospective intervention and is not represented as completed.

The conceptual layer asks a different question: what structure would connect difference detection, latent reconstruction, action selection, and consciousness-relevant access without confusing a metaphor with a measurement? It may motivate variables, alternatives, and future experiments, but it cannot alter the frozen Stage A target, folds, comparator ladder, SESOI, or decision rule after outcomes are inspected. No philosophical premise is allowed to function as evidence for a neural effect, and no neural decoding result is allowed to function as proof of a metaphysical claim.

2. Minimal Reconstruction Model

Let C_i be the cue-instructed grip condition on trial i , with values precision grip (PG) or side grip (SG). Let Y_i denote the multichannel neural observation in the pre-GO delay window. A task-relative render is the calibrated out-of-fold probability

$$q_i = P(C_i = PG \mid Z_i),$$

where Z_i is a declared feature set. The word *render* means a task-relative estimate available to a decoder or downstream controller. It does not mean a phenomenal image.

For electrode j , beta-band analytic activity is

$$x_j(t) = a_j(t) \exp(i \phi_j(t)), \\ \phi_j(t) = \arg x_j^{(13-30 \text{ Hz})}(t).$$

For undirected neighboring electrodes (j,k) , define the wrapped phase difference

$$\Delta \phi_{jk}(t) = \arg \exp(i[\phi_k(t) - \phi_j(t)]).$$

This quantity does not by itself assign causal direction. At each time point, fit the wrapped spatial plane

$$\phi_j(t) \approx \alpha_t + k_x(t) x_j + k_y(t) y_j \pmod{2\pi}.$$

The fitted gradient is $g_t = (k_x(t), k_y(t))$. A continuous lawful-gradient score is

$$L_t = |(1/N) \sum_j \exp(i[\phi_j(t) - \alpha_t - g_t \cdot r_j])|.$$

TNR features include gradient magnitude and direction, circular residual, local curl / divergence-like summaries on the array graph, temporal stability, and nearest-neighbor phase differences. No confirmatory conclusion depends on a binary wave / non-wave threshold.

3. Frozen Dataset And Task

3.1 Immutable data route

The source dataset is Brochier et al., *Massively parallel recordings in macaque motor cortex during an instructed delayed reach-to-grasp task*:

- data DOI: 10.12751/g-node.f83565;
- data repository: https://gin.g-node.org/INT/multielectrode_grasp;
- frozen repository commit: a6d508be099c41b4047778bc2de55ac216f4e673;
- article DOI: 10.1038/sdata.2018.55;
- data license: CC BY 4.0;
- repository code license: BSD 3-Clause.

The repository's tag 1.1.0 resolves to commit 24cd5cae3ae79066ca37844cab931d04dcad977, but that tagged tree does **not** contain the later datasets_nix exports. This protocol therefore freezes the exact later commit above rather than combining the tag identity with files it does not contain.

The reduced 1-kHz NIX files at the frozen commit are:

Subject	File	Annex identity
monkey L	datasets_nix/l101210-001_no_raw.nix	MD5 f20162beefd2af1077325c445035a9d0; 882,371,719 bytes

Subject	File	Annex identity
monkey N	datasets_nix/i140703-001_no_raw.nix	MD5 5c5e443f3b53361a1bed3ec1 bcad1b0b; 779,109,880 bytes

The files are not included in this manuscript package. Their local download hashes must be checked against the annex identities and recorded before analysis.

3.2 Development and replication roles

- development animal: monkey L (i101210-001);
- locked second-animal robustness subject: monkey N (i140703-001);
- target: binary cue-instructed grip condition, PG versus SG;
- primary window: GO-ON - 900 ms through GO-ON - 100 ms;
- primary band: 13-30 Hz;
- primary unit: one prediction and one loss per trial;
- inclusion: correct trials with complete CUE-OFF and GO-ON events and dataset-QC-passing LFP;
- exclusion decisions are label-blind and recorded before grip-conditioned scoring.

All transforms, feature definitions, baseline selection, model capacity, hyperparameter-selection rules, surrogates, and decision thresholds are developed on monkey L. Monkey N's condition-specific outcomes are not inspected until that configuration is frozen. Monkey N is a locked second-animal robustness test, not a direct anatomical replication: monkey L spans M1/PMd whereas monkey N is more lateral and includes PMv. Array coordinate orientation and hemisphere conventions must be frozen in the executable lock.

4. Signal Preparation And Feature Reliability

4.1 Primary pipeline

1. Load the reduced NIX recording and odML / Neo annotations.
2. Preserve the physical 10-by-10 Utah-array coordinates and four inactive positions.
3. Apply the dataset's channel / trial quality annotations without viewing label-conditioned outcomes.
4. Run the primary estimator causally on each continuous recording in acquisition order; no sample later than time t may contribute to the phase estimate reported at t .
5. End all confirmatory input 100 ms before GO-ON. The primary pipeline must embargo the estimator's declared latency and filter state so no sample after the cutoff, fold boundary, or trial boundary can enter a confirmatory feature.
6. Estimate 13-30 Hz phase with a strictly past-only state-space oscillator whose transition, observation noise, initialization, latency, missing-sample rule, and cutoff handling are frozen in the executable lock. A centered Butterworth-plus-Hilbert estimate is retrospective sensitivity analysis only and may never support the primary pre-GO claim.
7. Fit all centering, scaling, PCA, feature-selection, offset, and model parameters inside training folds only.

4.2 Sensitivity set

The bounded sensitivity set crosses:

- original reference, common-average reference, and 2-D surface Laplacian;
- the primary strictly causal state-space estimator and retrospective offline-Hilbert sensitivity;
- phase-quality thresholds at the training-fold 10th, 20th, and 30th amplitude percentiles.

The primary threshold is the 20th percentile. The threshold is fit per channel from training-fold samples only. The resulting time-point validity mask is supplied identically to every comparator, retained samples are reported by condition without outcome-based tuning, and a no-threshold sensitivity is mandatory. The confirmatory conclusion fails the robustness gate if its sign reverses across this bounded set.

5. Comparator Ladder And Capacity Contract

Every model receives the same outer folds, preprocessing access, optimizer budget, and locked capacity-selection rule.

Model	Feature family
M0	intercept and training-fold class prevalence
M1-rate	spike / event rate, broadband and cue-evoked summaries, trial-order nuisance
M2-power	M1 plus sitewise beta amplitude and power
M3-sync	M2 plus scalar PLV / coherence with no spatial gradient
M4-dynamics	non-oscillatory linear state / history features
M5-linear-LFP	unengineered 1-kHz LFP sequence plus geometry and history, compressed within fold and decoded linearly
M5b-nonlinear-LFP	the same unengineered 1-kHz LFP, mask, geometry, and history through an equal-compute causal temporal model capable of learning filtering, pairwise interactions, and temporal structure without supplied phase features
M6-TNR	the same input access plus declared phase-gradient, wave-geometry, and edge-relative phase features

The linear confirmatory family uses elastic-net logistic regression. Training-fold-only PCA chooses q from $\{8,16,24,32\}$ on monkey L. Equal PCA dimension is not treated as equal representational capacity. The nonlinear M5b comparator and M6 receive the same input samples, validity mask, training-example budget, hyperparameter-evaluation count, early-stopping rule, and maximum multiply-accumulate budget. Their exact architectures, initialization seeds, penalty grids, and optimization rules remain release blockers until frozen machine-readably. The selected q , capacity rule, and baseline champion are locked before monkey N.

6. Estimands, Nulls, And Decision Rule

Let $L(M)$ be out-of-fold binary log loss in bits per trial. Let M_{champion} be the strongest non-TNR comparator selected on monkey L under the locked rule. The primary paired increment is

$$\Delta_{\text{TNR}} = L(M_{\text{champion}}) - L(M6\text{-TNR}).$$

Positive values favor the phase-structured model. The superiority null is

$$H0_{\text{meaningful}}: \Delta_{\text{TNR}} \leq 0.02 \text{ bits / trial}.$$

The smallest effect of interest is 0.02 bits / trial. Meaningful support on monkey N requires:

1. Δ_{TNR} is at least 0.02 bits / trial;
2. a one-sided 95% full-pipeline block-bootstrap lower bound is above 0.02 bits / trial; and
3. the monkey-L development estimate has the same sign.

The equivalence null is

H_eq: $|\Delta_{\text{TNR}}| \geq 0.02$.

If the 90% confidence interval is entirely inside $[-0.02, +0.02]$, the meaningful representation- utility claim fails for this preparation. Positive estimates whose lower bound is above zero but not above 0.02 are reported as positive-but-sub-SESOI, not meaningful support. The other fixed outcomes are equivalent, inconclusive, or harmful.

Secondary estimands are held-out AUROC, balanced accuracy, next-100-ms LFP Gaussian log score, post-GO reaction-time prediction, and effect stability across the sensitivity set. Error prediction is removed because the primary cohort excludes error trials; any future error endpoint requires a separate all-trials cohort and lock. The log-score increment is not called literal mutual information unless calibration and density conditions are separately established.

7. Null And Surrogate Ensemble

Each null declares what it is intended to preserve and destroy. Those preservation claims must be verified numerically on the realized surrogate before interpretation, and the complete preprocessing, selection, and fitting pipeline is rerun for every replicate.

1. **Label permutation within contiguous temporal quintiles:** preserves slow drift and class counts; destroys label association.
2. **Whole-trial multichannel phase-map shuffle within temporal quintile:** preserves each map's power and spatial organization; destroys trial-label alignment.
3. **Independent per-channel Fourier phase scrambling:** preserves channel amplitude spectra; destroys cross-site phase relations.
4. **Independent circular shifts by preregistered lags longer than two beta cycles:** preserves autocorrelation and power; destroys inter-site and event alignment.
5. **Distance / degree-constrained array-coordinate permutation:** preserves signals and graph density; destroys anatomical embedding.
6. **Edge / preferred-offset shuffle:** preserves phase marginals; destroys the proposed stitching relation.
7. **Cue-locked evoked waveform and raw-sequence champion:** tests whether engineered phase features merely repackage evoked activity already available in the raw signal.
8. **Cycle-shape and acquisition controls:** test nonsinusoidal harmonics, aperiodic slope, mains contamination, sampling, and filter-edge artifacts.

The executable lock must additionally freeze the shuffle unit and strata; FFT window and padding; circular-shift lag distribution; coordinate-permutation group; random seeds; number of replicates; and model-refit rule. Until those are machine-readable, this is a pre-registration specification, not a signed analysis lock. Max-statistic permutation inference controls the surrogate family. Exploratory secondary endpoints use Benjamini-Hochberg false-discovery-rate correction. There is one confirmatory band, window, target, and primary model contrast.

8. Splits, Uncertainty, And Power

Monkey L uses five contiguous blocked outer folds with blocked nested inner cross-validation. Monkey N uses five contiguous blocked cross-fitting folds with the locked pipeline. Only out-of-fold trial predictions enter scoring. The complete preprocessing, hyperparameter selection, training, and scoring pipeline is rerun inside every bootstrap or permutation replicate; resampling a stored loss sequence alone is forbidden because outer training sets overlap. The primary moving-block length is 20 consecutive eligible trials, with 10- and

40-trial sensitivity analyses. Inference is conditional on these two recordings and is not a population-level macaque estimate.

This is a fixed-size public dataset. Before opening monkey N's grip-conditioned outputs:

1. use monkey L's paired out-of-fold loss sequence;
2. run at least 2,000 moving-block simulations at effects from 0 through 0.10 bits / trial;
3. report estimated power at the fixed 0.02 bits / trial effect; and
4. report the minimum detectable effect at 80% power.

If monkey N cannot supply 80% simulated power at the smallest effect of interest, Stage A remains a feasibility / estimation study regardless of a nominal p-value.

9. Stage A Execution Contract

The analysis is complete only when a clean environment can reproduce:

- a data manifest containing repository commit, file names, annex identities, downloaded hashes, trial counts, and QC exclusions;
- a configuration file with no outcome-selected confirmatory parameter and exact feature algorithms, graph edges, circular / spatial formulas, temporal summaries, model architectures, grids, optimizers, seeds, estimator state / latency rules, surrogate construction, refitting, and bootstrap rules;
- deterministic outer / inner fold assignments;
- out-of-fold predictions for every included trial and model;
- the complete surrogate ensemble;
- the power replay and uncertainty intervals;
- an all-runs manifest including failed and exploratory runs; and
- a machine-readable results file whose scope says `physical_intervention=false` and `consciousness_test=false`.

No confirmatory wording is written from an in-sample fit or from monkey N before the lock file, software environment, and immutable trial / fold assignment tables are signed and hashed. The present protocol still does not claim that the future empirical execution lock is complete.

10. Stage B Prospective Phase Perturbation

Stage B is a future causal protocol requiring separate ethics, safety, equipment, and acquisition approval. It should:

1. localize a receiver-specific beta phase-transfer curve independently;
2. randomize trials among open-phase, closed-phase, amplitude-only, and sham conditions;
3. match carrier frequency, pulse count, charge, energy, and cue timing;
4. use receiver spike / evoked-response transfer in a fixed 10-80 ms window as the primary endpoint;
5. treat task error and reaction time as secondary endpoints; and
6. test rescue by shifting receiver integration state and predicting a reversal of the preferred phase.

The primary causal estimand would be

$$\text{ATE}_{\phi} = E[Y \mid \text{do}\delta_{\text{open}}] - E[Y \mid \text{do}\delta_{\text{closed}}].$$

The provisional planning floor is 100 analyzable trials per condition, retained only if a pre-acquisition clustered simulation verifies at least 80% power for the chosen effect. Computational phase scrambling and feature deletion in Stage A are ablations, not biological interventions.

11. Falsification Policy

The Stage A claim fails for this preparation if any of the following holds:

1. monkey N places Delta_TNR inside the ± 0.02 -bit equivalence interval;
2. M5-linear-LFP or M5b-nonlinear-LFP matches or exceeds M6-TNR under the locked compute rule;
3. the increment vanishes after evoked-waveform, power, common-input, or spatial-embedding controls;
4. the effect changes sign across the bounded reference / phase-estimator sensitivity set;
5. phase variables fail the preregistered reliability gate;
6. only a post hoc band, time window, feature subset, or animal works; or
7. the locked pipeline cannot be reproduced from the frozen data and configuration.

A Stage B null selective phase effect would reject the proposed phase-specific causal transfer in that preparation. Neither Stage A nor Stage B alone adjudicates all oscillatory routing, all neural reconstruction, or consciousness.

12. Relation To SAN, NAPOT, And Micah's Genealogy

TNR is an empirical bridge extracted from the longer Self-Aware Networks (SAN) and Neural Array Projection Oscillation Tomography (NAPOT) genealogy. The documented sequence is not compressed into one retrospective date:

- **2011-2012:** predictive self / world modeling, recurrent feedback, neural computation, and slower / faster oscillatory structure appear in the early SAN dialogues;
- **2017:** the Neural Lace and 3-D-shape discussions make the spatial-predictive and multisensory read / write bridge explicit;
- **2022:** the GitHub corpus names NAPOT and develops distributed projection, waveform, phase, reader-relative, and tomography claims;
- **2024 onward:** later books and papers organize those atoms into explicit comparator and falsification programs.

The paper-relevant source ladder is more precise than those four summary bullets:

Stage	Date and date type	Bounded TCNR-relevant contribution	Provenance boundary
Early stereographic-camera note	May 13, 2011 owner / internal date; first immutable Git fixation June 12, 2022	Distributed “pixels” contribute partial scene information; statistical relations support inference and prediction; senses combine into a continuously updated multisensory picture.	The 2011 date is preserved as an internal authorial date, not represented as a 2011 public Git date. The immutable public artifact is the 2022 commit [24].
Public projector dialogue	August 23-24, 2012 service / public timestamp	The mind is described as a projector that predicts the world and supplies much of the experienced detail internally.	Functional ancestor of predictive neural rendering; it does not contain the mature TCNR operator [25].

Stage	Date and date type	Bounded TCNR-relevant contribution	Provenance boundary
Neural Lace Podcast	April 11, 2017 public timestamp	Spatial computation, a time-varying three-dimensional ionic matrix, feedback, prediction, and a brain-network protocol question are joined in one public discussion.	Public functional ancestor; it does not backdate NAPOT, PWD, or TCNR terminology [26].
NAPOT Git and public explainers	June-September 2022 immutable commits and service / public timestamps	Receive-transform-project arrays, distributed observation, phase-sensitive rendering, projection-slice and volumetric reconstruction analogies, memory scaling, and sensorimotor updating become an explicit joined program.	This is the mature public SAN / NAPOT source layer; biological validation remains separate [27,28].
Book consolidation	October 2024 published book and public Git records	Molecular, synaptic, dendritic, oscillatory, rendering, and action levels are consolidated into one sustained SAN account.	Published synthesis, not an empirical validation of TCNR [29].
TCNR4	October 19, 2025 embedded artifact date; November 15, 2025 immutable public Git fixation	The render operator, TCI proposal, six-condition gate, cross-substrate tests, SIT bridge, Telos, and optional Limit Mind are stated as one TCNR program.	Artifact date and public-fixation date are distinct. The exact 39-page PDF is hash-bound [23].
Draft 7 successor	July 27, 2026 current synthesis	The empirical protocol, historical conceptual program, corrections, failure conditions, and provenance ladder are preserved in one lossless but evidentially separated manuscript.	Current synthesis; it does not move any earlier date or report an empirical result.

The historical priority record and the scientific test have different jobs. The genealogy establishes what Micah publicly described and when. Stage A tests one present-day operational residual. A negative result would not erase the dated record; a dated record would not convert a negative experiment into a positive one.

An earlier full manuscript in this line, *TCNR4*, is the direct documentary predecessor to the present protocol [23]. Its embedded PDF metadata records creation on October 19, 2025. The exact 39-page artifact was fixed publicly in Git on November 15, 2025, in commit `eab1ea65c3abd488c2a33e6bf4f420c32951fcc3`. These facts establish artifact provenance and public availability. They do not independently validate TCNR’s biological, computational, consciousness-related, cross-substrate, or metaphysical hypotheses.

Part II. Firewalled Conceptual Bridge

13. From Difference To Pattern

The starting intuition is that a pattern is not one isolated element. It is a structured set of differences and relations among elements. Let the elements at time t be vertices $V_t = \{x_1, \dots, x_n\}$. For a declared measurement domain, define a relational difference

$$\Delta_{ij,t} = d(x_{i,t}, x_{j,t}), \quad (1)$$

where d may be Euclidean, circular, graph-geodesic, probabilistic, semantic, or another predeclared metric. A pattern candidate is then

$$P_t = (V_t, E_t, \Delta_t, C_t), \quad (2)$$

where E_t specifies which relations are eligible and C_t contains a bounded coherence or compatibility statistic defined for that domain. This notation does not say that every difference is neural phase or quantum coherence. It says only that isolated values become a relational object when their comparison rule, eligible edges, and measurement scale are declared.

Information theory already warns against equating a vivid pattern with information in the abstract. Information is defined relative to a distribution and an uncertainty reduction, not by aesthetic regularity alone [7]. Two coincident marks can be informative if they change the receiver's probability model, but coincidence does not automatically equal one bit. Likewise, a relation can be useful without being oscillatory, and oscillatory coherence can be high while carrying no task-specific information. TCNR therefore requires separate held-out estimates for discriminability, prediction, coherence, and action consequence.

The reconstruction map is

$$R(P_t) \rightarrow \hat{s}_t, \quad (3)$$

where $\hat{s}_t$ is a latent state or calibrated task-relative estimate. A part need not resemble the recovered whole. Many different projections can jointly constrain a latent object even when no single measurement contains the object in finished form. This is the defensible content of the jigsaw and tomography analogies. Literal tomography requires an identified forward operator, nonredundant projections, observability or identifiability conditions, a reconstruction criterion, and an operator-specific error test. Without those conditions, *relational recurrent reconstruction* is the narrower term.

13.1 Receiver-relative observability and mixture identifiability

A physical difference need not be created by an observer in order to matter. The receiver-relative question is narrower: can the difference affect the observations or actions available to a declared receiver? Let a physical state X generate a measurement Y through

$$p(y \mid x), \quad \text{or} \quad y = H(x) + \epsilon. \quad (4)$$

Two physical states may differ while producing indistinguishable measurement distributions for one channel. Conversely, a channel may separate them reliably. TCNR therefore uses *receiver-relative* to mean accessible, discriminable, or causally usable through a specified observation-and-action interface. It does not mean that observation creates the underlying physical difference.

This boundary is also an inverse-problem constraint. If one observed composite satisfies

$$y = F(x_1, x_2), \quad (5)$$

and several pairs (x_1, x_2) generate the same y , the latent components are not identifiable from that observation alone. Recovering the components requires additional channels, interventions, independence or sparsity assumptions, a calibrated forward operator, or other constraints that can themselves be tested. A compelling composite percept is therefore evidence about the receiver’s render, not automatic evidence that the receiver has recovered the composite’s unique causal history. This is why TCNR evaluates task-relative prediction and perturbation while refusing to call an unconstrained latent decomposition literal tomography.

13.2 Four meanings of coherence

The word *coherence* is overloaded. TCNR must keep at least four meanings separate.

Domain	Operational meaning	Suitable observable	What cannot be inferred
Statistical dependence	Variables depart from a declared independence model.	Mutual information, likelihood ratio, covariance, conditional dependence.	Phase locking, causal direction, or consciousness.
Neural oscillatory coherence	Band-limited signals have a reproducible cross-spectral phase / amplitude relation at a declared scale.	Complex coherence, imaginary coherence, phase-locking value, directed model with controls.	Quantum coherence, semantic agreement, or sufficient integration.
Computational integration	Several inputs jointly improve a latent estimate or policy beyond their parts.	Out-of-sample synergy, ablation, reconstruction error, control gain.	A particular oscillatory implementation or subjective experience.
Quantum coherence	Off-diagonal structure relative to a specified basis survives in a density operator and supports interference under controlled dynamics.	Quantum-state or process tomography and resource-theoretic measures [11,12].	Macroscopic mental unity, neural phase synchrony, or scale invariance.

The same mathematical word can appear in these rows without making the physical mechanisms identical. A cross-scale theory needs an explicit translation map, dimensional consistency, and a test that could fail at each bridge.

A single temporal coincidence is not yet a coherence estimate. Statistical or oscillatory coherence requires repeated observations, a declared estimator, and comparison with a null that preserves the relevant marginal structure. Stable classical phase locking can arise from direct coupling, common input, filtering, reference choice, or shared task drive; it does not by itself establish causal direction or additional task information.

13.3 Surprise, clarity, meaning, and causal value

A rare event is not automatically meaningful, true, useful, or clear. **Conditional surprise**, or conditional self-information, can be written

$$I_t = -\log_2 p(y_t \mid y_{<t}, a_{<t}), \quad (6)$$

but high surprise can arise from noise, sensor failure, or an irrelevant outlier. Bayesian information gain instead measures the change between prior and posterior beliefs,

$$IG_t = D_{KL}[q(s_t \mid y_{\leq t}) \parallel q(s_t \mid y_{<t})], \quad (7)$$

and still does not guarantee semantic relevance or action value [17]. TCNR therefore separates:

- **surprise:** improbability under a model;
- **discriminability:** separability at a receiver;
- **information gain:** posterior change;
- **clarity:** estimator reliability or uncertainty;
- **semantic relevance:** relation to a task or representational variable; and
- **causal value:** change in prediction or action under intervention.

A phasic event counts as evidence for TCNR only when it improves a held-out endpoint or changes a declared receiver / action variable under the relevant controls. A bright spectral event or rare coincidence is not sufficient.

13.4 Pattern tokens, pattern types, and persistence

Claims that an object or technology is an evolving pattern can refer to several different things:

1. one physical token persists while its microstate changes;
2. a type recurs in many different physical tokens;
3. a cultural or technological lineage transmits constraints across tokens;
4. a system causally repairs or reproduces itself; or
5. an observer groups different tokens under one concept.

A lamp design recurring across centuries is mainly type-level and cultural continuity. It is not one physical state literally persisting, and it does not show intrinsic agency. This distinction is important for cross-substrate TCNR: structural recurrence, self-maintenance, reproduction, prediction, and counterfactual action are separate criteria.

13.5 Coherence-matrix determinant correction

The original TCNR source treated positive $\log \det(C + \epsilon I)$ as a generic coherence reward. That direction is not valid without a specific interpretation. For a two-variable normalized correlation matrix

$$C = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}, \quad \det C = 1 - \rho^2, \quad (8)$$

the determinant decreases as $|\rho|$ increases. A positive log-determinant can therefore reward independence or spectral volume rather than synchrony. The present protocol does not use that term as its primary endpoint. Any future multivariate summary must declare whether it measures dependence, diversity, effective dimensionality, or precision; establish its monotonic direction on simulated cases; and compare it with direct coherence / phase-locking and predictive endpoints. No scalar determinant is a consciousness measure.

14. From Pattern To Prediction And Action

Recognition alone does not close the TCNR loop. A reconstructed state matters computationally when it changes prediction, policy, or control. Let b_t be the agent or controller state, a_t an action, and y_{t+1} the next observation. A minimal loop is

$$\hat{s}_t = R(P_t), \quad a_t \sim \pi(a \mid \hat{s}_t, b_t), \quad P_{t+1} = U(P_t, a_t, y_{t+1}). \quad (9)$$

This yields the causal sequence

$$\begin{aligned}
& \text{differences} \\
& \rightarrow \text{relational pattern} \\
& \rightarrow \text{latent estimate} \\
& \rightarrow \text{predicted consequence} \\
& \rightarrow \text{selected action} \\
& \rightarrow \text{feedback} \\
& \rightarrow \text{updated pattern.}
\end{aligned} \tag{10}$$

Active-inference and predictive-processing frameworks provide important prior architectures for linking generative models, perception, planning, proprioception, and action [8-10]. Their broad explanatory reach is not itself proof of TCNR. Recent reviews emphasize that empirical adequacy requires comparisons against alternative models and qualitatively distinct predictions [9,10]. The TCNR residual is narrower: after the same raw data, history, capacity, and compute are supplied to the strongest comparator, do declared relational phase-geometry variables improve prediction or control, and does selective phase perturbation change receiver transfer?

Intention should not be reduced to reflex. A state-dependent response may be intentional only if the system represents or realizes alternatives in a way that affects policy. TCNR's counterfactual capacity condition is meant to prevent a fixed oscillator or stimulus-response chain from being called deliberative merely because it is coherent. Candidate operationalizations include policy diversity, local controllability, or behavior under counterfactual cue changes. None is yet a validated consciousness criterion.

15. Consciousness Science: What TCNR Can And Cannot Test

Contemporary consciousness theories address different targets: phenomenal character, conscious access, report, metacognition, global availability, recurrent processing, higher-order representation, predictive self-modeling, and causal integration. They are not interchangeable [13-15]. TCNR should therefore avoid a single scalar *consciousness index* until a target construct, measurement model, and external validation are established.

The original TCNR source proposed a weighted TCI combining predictive mutual information, coherence, counterfactual capacity, and prediction error. That formula is retained as historical design material, not as a validated measure. A weighted sum can hide compensation: high coherence might numerically offset absent counterfactual behavior even if both were claimed necessary. It can also mix estimators with different reliability, units, and causal status. The present paper replaces that scalar with a **claim vector**:

$$X_t = (I_{pred}, E_{recon}, C_{rel}, K_{cf}, G_{access}, S_{self}), \tag{11}$$

where:

- I_{pred} is held-out predictive information or a proper-score improvement;
- E_{recon} is operator-specific reconstruction error;
- C_{rel} is a declared relational / coherence statistic;
- K_{cf} is counterfactual policy or controllability capacity;
- G_{access} is a separately measured access, report, or task-availability endpoint; and

- S_{self} is an independently defined self / body / agency endpoint when relevant.

No component is silently imputed, and no weighted total is called consciousness. Different studies may test different arrows. Stage A in this manuscript tests only a representation-utility residual for a cue-instructed grip condition. It does not measure G_{access} or S_{self} and is not a consciousness experiment.

A future consciousness-relevant TCNR program would need at least:

1. a declared construct such as access, metacognition, or state transition;
2. matched attention, memory, decision, report, and motor controls;
3. a phase-structure model compared with equal-input alternatives;
4. a causal intervention with a construct-specific endpoint;
5. negative conditions in which coherence exists without the target construct; and
6. replication across tasks or measurement contexts.

Even success would support a mechanism contributing to the declared construct, not a proof that TCNR is necessary or sufficient for phenomenal experience.

15.1 Intersubjective render convergence

Different people can report similar experiences or memories because they share sensory evidence, language, cultural priors, leading questions, source-monitoring errors, or post-event discussion. Classic false-memory and memory-conformity experiments show that structured convergence can be produced without a shared percept or altered history [18,19]. TCNR can turn this into a prospective social reconstruction test:

1. elicit independent reports before discussion;
2. manipulate whether participants receive common versus distinct cues;
3. vary post-event social exposure;
4. model semantic attraction and source confidence;
5. compare individual and group-level latent reconstructions; and
6. test whether convergence tracks shared evidence and transmission rather than an unexplained common internal state.

The permitted term is **intersubjective report convergence** or **collective false memory** when the content is false. Similar reports do not establish a literally shared hallucination, alternate timelines, collapsed universes, or retroactive changes to the past.

15.2 Conscious time and computational event rate

The intuition that more computation per physical second could change experienced time is a useful question, but it divides into several endpoints:

- physical elapsed time;
- neural or computational event rate;
- online duration judgment;
- temporal resolution or discrimination;
- retrospective remembered event density; and
- phenomenal duration.

These can dissociate [20]. An external processor could increase prediction depth or memory density without proportionally lengthening online experienced duration. A valid experiment would manipulate processing or event density while separately measuring prospective duration judgment, retrospective duration, temporal

discrimination, memory count, performance, arousal, and report strategy. TCNR makes no present claim that computation rate is consciousness rate.

15.3 Historical TCI and six-condition program

The source record must not disappear merely because its evidential ceiling changed. TCNR4 proposed a Tomographic Consciousness Index (TCI) that combines four null-standardized components: predictive information, a band-limited coherence summary, counterfactual capacity, and negated prediction error. It also proposed a six-part gate covering band-limited coherence, lawful traveling-wave routing, closed feedback, memory / hysteresis, counterfactual agency, and dynamic inference [23]. In TCNR4 these conditions were presented as jointly necessary rather than sufficient; a later metaphysical companion used stronger language. Draft 7 preserves the original proposal while separating three statuses:

1. **historical specification:** the formula, component list, nulls, and gate belong to the dated TCNR4 record;
2. **current measurement program:** each component can be estimated and tested separately through the claim vector in Section 15; and
3. **unvalidated interpretation:** neither $\text{TCI} > 0$ nor passage of the six-part gate is presently established as a criterion for consciousness.

The log-determinant correction in Section 13.5 is therefore an additive repair, not deletion of the coherence term. A future TCI successor may use a directionally validated dependence, diversity, or effective-dimensionality statistic, but it must calibrate the construct against positive and negative consciousness-relevant conditions and must prevent one component from numerically hiding failure of another. The historical TCI, its benchmark idea, and its adversarial nulls remain available as a research program.

16. Physics Bridge: Local, Conditional, And Testable

The physics language in TCNR is motivated by a legitimate observation: neural signals are physical, phase and frequency are physical variables, and reconstruction ultimately occurs in a material substrate. The bridge becomes unsafe when similar words are treated as identities. Neural oscillatory coherence is not quantum coherence. Classical simultaneity is frame-dependent rather than an observer-free statement that two distant events happen at the same time. A local field interaction cannot be inferred from matching patterns at arbitrary separation.

16.1 Scale invariance is not scale identity

A relation may have the same dimensionless form at several scales while its carrier, noise, coupling, energy, and causal range differ. If a model claims scale invariance, it should specify a transformation S_λ and the quantities preserved:

$$\mathcal{M}[S_\lambda(x)] = \mathcal{T}_\lambda(\mathcal{M}[x]). \quad (12)$$

The claim is empty until S_λ , $\mathcal{T}_\lambda$, and the error tolerance are defined. Locality is a separate condition: interaction strength, delay, or influence must be a function of physical or graph distance under a declared propagation law. Two far-separated events can be statistically coincident without exchanging a signal or forming a physical unit.

16.2 Quantum coherence and decoherence

In quantum theory, coherence is basis-relative and described through a density operator; resource- theoretic measures and allowable operations make the definition explicit [11,12]. Decoherence is an open-system process in which environmental interaction suppresses interference in a selected basis; it is not a synonym for disagreement, randomness, neural desynchronization, or a pattern boundary. The TCNR difference graph may use *coherence* and *decoherence* as bounded relational labels only if the measurement domain is named. Quantum terminology requires an actual quantum state / process model and observables that distinguish it from a classical stochastic account.

Classical synchronization is also not quantum entanglement. Matching frequencies, a stable relative phase, or correlated responses can be represented by classical coupled oscillators. Quantum entanglement concerns nonseparability of a joint quantum state and requires an actual quantum-system description plus a valid entanglement witness, state / process reconstruction, or another certified criterion [22]. Bell-inequality violation is one stronger device-independent route in suitable experiments, not a prerequisite for every entanglement claim. Persistence of classical phase locking until perturbation does not make the signals entangled, and loss of neural synchrony is not quantum disentanglement.

16.3 Spacetime, computation, and information

The philosophical proposition that spacetime is computational can be separated into at least three claims:

1. physical evolution can be represented computationally;
2. the universe literally implements a computation; and
3. information or computation is ontologically fundamental.

The first is methodologically ordinary. The second requires a mapping from physical states to computational states and a nontrivial account of implementation. The third is metaphysical. TCNR does not establish claims 2 or 3. A neural reconstruction result supplies no evidence that spacetime itself is a mind or computer.

The more defensible bridge is structural: physics, neuroscience, and artificial systems can each be described using states, relations, transformations, observables, and interventions. Structural similarity can motivate a formal translation, but only empirical adequacy and bridge-specific tests justify identification. This stance is compatible with structural-realist questions while remaining neutral about whether relations or objects are ontologically fundamental [16].

16.4 Time asymmetry and thermodynamic language

The original source also linked coherence flux, memory, and a time-density motif to the arrow of time. A defensible stochastic-thermodynamic bridge begins with forward and reversed path probabilities. Under suitable conditions,

$$\Sigma = D_{KL}(P_F||P_R) \geq 0 \quad (13)$$

quantifies path-space irreversibility or entropy production [21]. The time derivative of an arbitrary KL divergence is not automatically nonnegative and is not automatically a physical entropy-production rate. Boundary conditions, coarse graining, reservoirs, units, and the reverse process must be defined.

In TCNR, **time density** may be used operationally for an event, update, or memory-encoding rate per declared clock interval. It is not identified with relativistic proper time or a gravitational source unless a separate dimensionally consistent theory supplies and tests that bridge. Neural hysteresis can support a memory trace without proving a cosmological arrow of time.

16.5 Historical SIT, cosmic, teleonomic, and metaphysical bridges

TCNR4 also placed the local render loop inside a larger Super Information Theory (SIT) vocabulary: coherence fields, an informational-event-rate or “time-density” variable, informational energy, memory as hysteresis, and a forward arrow associated with dissipative history [23]. It then introduced a teleonomic objective - called Telos - in which a renderer’s policy trades integrated prediction and valued outcomes against unnecessary dissipation. Finally, it described an optional cosmic limit construction: maximize a corrected TCNR objective over physically admissible render-stitch systems, examine limiting sequences, and ask whether a maximally capable renderer could support a separate Principle-of-Sufficient-Reason or modal argument.

Those propositions are preserved here because they are part of Micah Blumberg’s authored argument, but they do not inherit evidence from Stage A. Four different research questions must be kept distinct:

1. **computational teleonomy:** does a declared objective explain or predict adaptive policy under resource constraints?
2. **thermodynamic realization:** can objective terms be mapped to measured work, entropy production, or dissipation with consistent units and boundary conditions?
3. **cosmic generalization:** is there an identified physical state, observation map, renderer, and intervention at the proposed scale?
4. **metaphysical inference:** do independent premises such as sufficient reason or modal logic support a conclusion after the empirical claims are removed?

The local neuroscience protocol answers none of questions 2-4 and only motivates a bounded version of question 1. The SIT, Telos, cosmic-renderer, and metaphysical constructions therefore remain historical TCNR4 propositions and companion-paper routes. They are not retracted, silently erased, or represented as results of the macaque analysis.

17. Cross-Scale And Cross-Substrate Program

TCNR is not licensed to move from cortex to plants, biofilms, machines, or cosmology by analogy alone. A cross-substrate claim requires the same typed causal interface:

$$\begin{aligned} & \text{partial observations} \\ & \rightarrow \text{declared relational transform} \\ & \rightarrow \text{latent estimate} \\ & \rightarrow \text{policy / action} \\ & \rightarrow \text{feedback and adaptation.} \end{aligned} \tag{14}$$

For each substrate, the study must name:

- the state variables and observation operator;
- the receiver or integrator and its physical scale;
- the relational / coherence estimator and nulls;
- the action or control endpoint;
- the counterfactual intervention;
- the adaptation timescale; and
- an equal-input alternative model.

The cross-scale hypothesis is supported only when the typed operation and discriminating residual survive in more than one substrate. It is rejected or narrowed when the apparent match depends on metaphor, inconsistent units, outcome-selected bands, or a missing action / feedback arrow. A universe- level application is a philosophical limit question until those fields and interventions are defined; it is not an empirical conclusion of the macaque protocol.

TCNR4 specifically proposed plant, biofilm, artificial-oscillator, clinical, and perturbational benchmark routes [23]. Draft 7 preserves those routes as prospective hypotheses:

- a substrate should expose partial observations rather than only a finished global state;
- a measurable relational transform should improve prediction or control beyond equal-input alternatives;
- phase, amplitude, power, common drive, geometry, and generic model capacity should be separated;
- perturbing the proposed relation at matched energy should alter a receiver or behavioral endpoint; and
- high synchrony without repertoire, adaptation, or task benefit should count as an adversarial false positive.

No default extension is granted. Each substrate must earn the TCNR label by passing its own measurement, comparator, perturbation, and failure conditions. This converts the original cross-substrate assertion into a preserved and falsifiable research program rather than deleting it.

18. Non-Claims And Limitations

This protocol does not claim that:

- cue-instructed grip-condition decoding is consciousness or a cue-independent maintained plan;
- coherence or beta phase is universally necessary;
- a positive decoder proves a volumetric neural image;
- a planar gradient uniquely identifies a traveling wave;
- phase differences assign causal edge direction;
- motor-cortex findings generalize to vision without a separate dataset;
- plants, biofilms, or artificial systems instantiate the same mechanism;
- TNR validates all of SAN, NAPOT, SIT, or QGTCD; or
- a null result in two animals disproves phase-dependent computation everywhere;
- neural phase coherence is quantum coherence;
- classical coincidence, matched frequency, or phase locking is quantum entanglement;
- receiver-relative observability means that an observer creates the underlying physical difference;
- one composite measurement uniquely identifies its latent causes without a forward model or added constraints;
- statistical coincidence implies physical interaction or causal influence;
- scale-invariant notation makes mechanisms identical across scales;
- spacetime is empirically shown to be a computer, mind, or consciousness substrate; or
- the conceptual difference-pattern-action ladder validates a consciousness index.

The two-animal sample limits population generalization, and their different M1/PMd versus lateral / PMv coverage prevents a direct anatomical-replication claim. The same animals cannot provide both model discovery and a large independent population estimate. The fixed public data may be underpowered at the declared smallest effect. Task cues, evoked responses, movement preparation, volume conduction, reference choice, cycle shape, and array geometry are serious alternatives addressed by controls but not metaphysically eliminated.

19. Data, Code, Ethics, And Release Status

The source experiment's ethics and animal-care approvals are reported by Brochier et al. This protocol performs secondary analysis of public data and introduces no new animal procedure. Stage B would require new approval.

No large data file is redistributed here. The exact repository commit and annex identities are recorded so an authorized user can retrieve the source from G-Node under CC BY 4.0. Analysis code, environment lock, downloaded-file hashes, and results will be released only after the pipeline is implemented, replayed, and reviewed.

This Draft 7 is a locally final-preprint-ready, adversarially integrated theory-and-protocol successor. Part I preserves the prospective neural test; Part II is a conceptual bridge with no empirical authority over that test. The paper contains no empirical result. The exact local package is suitable for an author-controlled preprint-version decision, but this manuscript does not itself perform a Zenodo deposit, OSF deposit, DOI action, upload, or other provider mutation.

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